

Project 02

AI Music & Audio Analyzer

Living Project Documentation — v0.2

Item	Current Value
Status	In Progress — Audio Capture & Basic Analysis Verified
Platform	Windows Laptop
Python	3.14.7
Environment	VS Code + PowerShell + Python venv
Hardware	Windows Laptop + Airdopes Joy Bluetooth Earbuds
Input	Airdopes Joy Hands-Free microphone
Output	Airdopes Joy Stereo
Analysis	Librosa + NumPy + Matplotlib

1. Project Definition

AI Music & Audio Analyzer is a Python project for real hardware audio capture, digital signal processing, visualization, music/audio feature extraction, and later AI interpretation. The Windows laptop is the processing platform and Airdopes Joy provides microphone input and playback.

2. Architecture

Airdopes Joy Microphone → Bluetooth → Windows Audio → Python sounddevice → WAV Capture → Librosa / NumPy / Matplotlib → Analysis → Reports / Dashboard

3. Completed Milestones

Step	Milestone	Status
1	Audio device detection	PASS
2	Airdopes microphone verification	PASS
3	Five-second microphone recording	PASS
4	WAV loading and basic analysis	PASS
5	Waveform visualization	PASS
6	Audio level analysis	PASS

4. Step 1 — Audio Device Detection

Airdopes Joy Hands-Free microphone was identified as device 2 with one input channel and a reported 44.1 kHz rate. Airdopes Joy Stereo was identified as device 4 with two output channels at 44.1 kHz.

5. Step 2 — Microphone Verification

The Airdopes Joy Hands-Free microphone was successfully accessed with Python sounddevice and produced a non-zero signal.

6. Step 3 — Airdopes Audio Recording

A five-second mono recording was captured from device 2 and saved as `audio\airdopes_test.wav`.

Parameter	Result
Device ID	2
Input Channels	1
Sample Rate	44,100 Hz
Duration	5.00 seconds
Samples	220,500
RMS Amplitude	0.031082
Peak Amplitude	0.423889

7. Step 4 — WAV Audio Analysis

The saved WAV file was loaded successfully with Librosa. The measurements matched the recording stage, confirming reliable file processing.

Sample Rate: 44,100 Hz | Channels: 1 | Samples: 220,500 | Duration: 5.00 sec

RMS: 0.031082 | Peak: 0.423889 | RMS: -30.15 dBFS | Peak: -7.45 dBFS

8. Step 5 — Waveform Visualization

The recorded signal was converted to a time-domain waveform with Librosa and Matplotlib. The graph was successfully saved as reports\airdopes_waveform.png.

9. Step 6 — Audio Level Analysis

A dedicated analyzer calculated amplitude statistics, RMS, peak, dBFS levels, crest factor, and RMS level over time.

Measurement	Result
Minimum Amplitude	-0.347687
Maximum Amplitude	0.423889
Mean Absolute Amplitude	0.010442
RMS Amplitude	0.031082
Peak Amplitude	0.423889
RMS Level	-30.15 dBFS
Peak Level	-7.45 dBFS
Crest Factor	13.638
Classification	Healthy
Graph	reports\airdopes_audio_level.png

10. Current Project Structure

Project-02-AI-Music-Audio-Analyzer/

■■■■ audio/ → airdopes_test.wav

■■■■ reports/ → airdopes_waveform.png, airdopes_audio_level.png

■■■■ src/ → audio_devices.py, test_airdopes_mic.py, record_airdopes.py, analyze_airdopes_audio.py, plot_airdopes_waveform.py, audio_level_analysis.py

■■■■ tests/ ■■■■ venv/ ■■■■ requirements.txt ■■■■ README.md

11. Testing Record

Test	Result
Device enumeration	PASS
Airdopes microphone detected	PASS
Airdopes stereo output detected	PASS
Microphone capture	PASS
WAV generation	PASS
WAV loading	PASS
Waveform generation	PASS
Audio level analysis	PASS
RMS graph generation	PASS
FFT / frequency spectrum	NEXT

12. Next Step

Step 7 — FFT / Frequency Spectrum Analysis using the same airdopes_test.wav recording. The frequency-domain spectrum and dominant frequency components will be calculated and visualized.

13. Version History

v0.1 — 19-Aug-2026: New documentation baseline created from the beginning; hardware setup, device detection, microphone verification and recording documented.

v0.2 — 19-Aug-2026: Steps 4–6 completed. WAV loading, waveform visualization, RMS/peak/dBFS analysis, amplitude statistics, crest factor and RMS level graph documented with actual measured results.

Only the new Project-02 history is used; previous report versions are intentionally excluded.